

James Wang - Portfolio

B.S. Mechanical Engineering at the **Georgia Institute of Technology**

December 2027 Graduation

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McLaren Formula One Portfolio

- July 2025 – July 2026
- Engineering Intern
- Woking, Surrey, UK



Conformal Front Wing Proximity Sensors

Situation:

There was an opportunity to test a front-wing development sensor package on track during a practice session within a one-week timeline. The design had to place sensors on the front wing while maintaining consistent sensor positioning and minimizing aerodynamic intrusion from both the mounts and wiring harness. The project was difficult because the mounts needed to conform to the front wing geometry, survive vibration and bump loading, minimize package size, and be installed accurately within a tightly constrained running schedule.

Task:

I was responsible for the mechanical design and delivery of the sensor mounting system from concept to installation. This included the sensor mount design, wire harness routing, assembly/trimming drawings, and bonding jigs.

Action:

Using 3DX, I designed conformal aerodynamic sensor mounts for the front wing surface while creating internal conformal features to house each proximity sensor. To minimize aerodynamic impact, I worked with the aerodynamic performance team to define low-intrusion wire harness routing behind the sensor mounts. I selected a 3D-printed SLS carbon-reinforced material because it offered high strength, a high-quality surface finish after vapor smoothing, and wide availability in the factory. Finally, I performed hand calculations using worst-case bump and vibration accelerations to size the bonding area required for each sensor.

Result:

The full package was delivered from concept to installation in one week and made it to track for a practice session. The sensor mounting blocks were produced and assembled, and the design worked successfully on the first running. The sensors showed no signs of disbonding over harsh curbs and bumps and provided the team with valuable development data. The project highlighted my ability to deliver safe, functional engineering hardware quickly under pressure.



Public FIA / F1 front wing example for context only.
Not representative of McLaren design, geometry, packaging, or implementation.
Source: FIA / Formula 1 public materials.

Active Aero Mechanism Linkage Pins

Situation:

For a new vehicle package, new linkage pins connecting an actuator to an active aero mechanism had to be designed. The pins were simple in appearance but critical to the safe operation of the car, since a failure in the linkage could compromise vehicle stability. The design was difficult because each pin also functioned as the inner race for a bearing, requiring high surface hardness, tight tolerance control, and a high-quality surface finish while still surviving worst-case fault loading conditions.

Task:

I was responsible for the full mechanical design and delivery of the two linkage pins. This included the CAD, hand calculations, material selection, heat treatment specification, tolerance stackup, drawing, manufacturing support within the factory, and feedback loop with the FEA team. The goal was to produce a pin design that could satisfy peak load and fatigue requirements, act as a bearing race without premature wear, and remain manufacturable after heat treatment and post-process grinding.

Action:

Using 3DX, I designed the linkage pins around the bearing and active aero mechanism interfaces. I performed shear and bearing stress calculations, as well as Hertzian contact stress calculations to determine whether the bearing loads would cause surface yielding. These calculations necessitated a heat-treated high-strength steel with high surface hardness. Because the pins needed to meet a very tight final diameter and surface finish after heat treatment, I accounted for heat treatment depth and post-process grinding in the tolerance stackup and drawing. I also worked in the feedback loop with the FEA team, making small geometry changes to satisfy fatigue and peak load simulations.

Result:

The linkage pins were successfully designed, manufactured, and used as reliability-critical hardware for the active aero mechanism. The project highlighted my ability to own critical hardware through the entire engineering design process.



Public FIA / F1 active-aero mechanism example for context only.
Not representative of McLaren design, geometry, packaging, or implementation.
Source: FIA / Formula 1 public materials.

Hydraulic Manifold Model and Drawing

Situation:

A high-pressure hydraulic manifold from a previous design iteration needed to be updated to a new car design. The existing manifold model was a rough starting point, but it needed to be refined into a robust, manufacturable, release-ready part. The manifold had highly complex geometry, interfaced with multiple parts in the assembly, required multiple angled holes and cross-bores, and had tight sealing interfaces that drove tight tolerances, complex tolerance stackups, and difficult inspection requirements.

Task:

I was responsible for refining the manifold CAD model and updating the manufacturing drawing. This included modifying the model for new features, adding an additional cross-bore, repositioning several holes and threads to meet minimum wall thickness requirements, updating the GD&T, and supporting manufacturing if there were questions or inspections failed. Because the part had hydraulic interfaces with many other parts, the drawing needed to clearly define the sealing interfaces, hole geometry, inspection strategy, and datum schemes required to make and inspect the part correctly.

Action:

Using 3DX, I updated the manifold CAD to fit new design requirements while maintaining a robust model structure for a complex hydraulic part. I checked minimum wall thickness using CAD and drawing tolerances, then repositioned holes and threads to satisfy those requirements. In addition, I added the required cross-bore while staying within the tight constraints of the surrounding assembly. I also produced a full multisheet GD&T drawing for the manifold. The drawing had to communicate multiple machining setups, angled holes, cross-bores, tight sealing interfaces, and many datums in a way that manufacturing and inspection teams could use. During manufacturing support, I reviewed a nonconformance by checking the model, drawing, and tolerance calculations against the manufactured part. After confirming the part was outside acceptable requirements, I rejected the part and prevented it from progressing into assembly.

Result:

I delivered an updated high-pressure hydraulic manifold model and drawing for the new car design and regulations. In addition, I prevented non-conforming hydraulic components from progressing into assembly. The project highlighted my ability to refine complex CAD models, understand high-pressure hydraulic design constraints, create detailed GD&T drawings for difficult 5-axis machined parts, and support manufacturing decisions with evidence and calculations.

Titanium Quick Disconnect Keep Plate

Situation:

Two titanium keep plates were needed to retain high-pressure quick-disconnect connectors on hydraulic manifolds for race hardware. Although the parts were simple in geometry, they were important because they provided assembly retention and helped prevent pressure-related ejection of the connectors.

Task:

I owned the design of two different keep plates from concept to manufacturing support. This included the CAD, hand calculations, material selection, drawing, and support during manufacturing. The keep plates needed to interface cleanly with the quick-disconnect collars and manifold surfaces, retain the connectors under worst-case pressure loading, and be simple to manufacture through laser cutting and machining.

Action:

Using 3DX, I designed the keep plates around the connector collars and manifold interfaces. I calculated the retention force from worst-case pressure loading using conservative assumptions, then simplified the geometry to perform yield checks and fastener checks by hand. I also worked through a quick feedback loop with the FEA team to confirm the design and remove unnecessary material for weight optimization.

I selected titanium because it provided excellent specific strength and was relatively cheap due to the existing machining and tooling infrastructure at McLaren. I also intentionally designed the keep plates asymmetrically so they could only install in the correct orientation.

Result:

The keep plates have worked successfully with no retention issues. During assembly, there was a question that arose when one part did not initially fit, but the asymmetric design made it clear that the keep plate was simply being oriented incorrectly. Flipping it allowed the part to fit as intended, preventing a potential incorrect assembly. The project highlighted my ability to treat reliability-critical hardware with the proper engineering depth while still optimizing for performance to save every single gram possible.

Fluid Tank Baffle

Situation:

A carbon composite internal baffle needed to be redesigned for a new fluid tank package. Its primary function was to control fluid movement, but there was also an opportunity to use the baffle as a local structural member to improve stiffness and reduce overall system mass.

Task:

I was responsible for modifying the existing baffle design into a release-ready component for the new package. This included CAD updates, baffle placement, cutout design, FEA feedback incorporation, and laminate manual updates.

Action:

Using 3DX, I repositioned the composite baffle and modified its cutouts to control fluid movement while reducing unnecessary mass. Because analysis bandwidth was limited, I based the layout on prior analysis results and engineering judgment from earlier designs, then used FEA feedback to improve stiffness in targeted regions. I adjusted the geometry so the baffle could still perform its primary fluid-control function while also contributing structurally to the surrounding assembly.

Result:

The redesigned baffle was produced and used successfully. The final design turned a required internal baffle into a component that also contributed structurally to the tank assembly and supported overall weight reduction. The project highlighted my ability to find secondary value in existing components to improve the system rather than only optimizing the part in isolation.

Rear Wing Panel - 3-Piece Mold, Pattern, and Laminate

Situation:

A rear wing panel for a new aerodynamic package needed to be made from a complex aerodynamic surface that also had local structural requirements. The surface itself was already defined, but turning it into composite hardware was difficult because the geometry included undercuts that made a simple two-piece pattern impractical. A pattern and mold approach needed to be designed that preserved the aerodynamic surface, maintained clean mating edges, and achieved acceptable draft angles.

Task:

I was responsible for turning the defined aerodynamic surface into a manufacturable composite part. This included the pattern design, mold design, flanges, split lines, bolt holes, alignment dowels, and laminate manual support. I mainly worked independently on the project, taking the defined aerodynamic surface and developing the pattern, mold, and laminate approach needed to produce the panel.

Action:

Using 3DX, I developed a three-piece pattern approach because the undercut geometry made a two-piece pattern impractical without compromising draft angles or mating edges. I designed the pattern split lines, flanges, bolt holes, and alignment dowels so the tooling could locate repeatably and come apart cleanly while preserving the final panel geometry.

The pattern was used to create a single-piece female carbon mold, allowing the final carbon panel to be produced with a clean surface finish and minimal thermal expansion mismatch between the mold and part during cure. I also supported the laminate manual by defining the general ply layout, trim regions, and local reinforcement areas for structural features.

Result:

The tooling and laminate approach worked correctly, producing a rear wing panel that satisfied the aerodynamic surface requirements while remaining manufacturable despite the undercut geometry. The project highlighted my ability to take complex aerodynamic surfaces and turn them into real composite hardware through pattern design, mold design, and laminate planning.

Remote Access Steering Rack Bleed Tool

Situation:

The steering rack on the car needed a better way to be bled during assembly and disassembly. The existing tool required deep access in the nose of the car, where packaging is extremely tight once the nose is assembled. A longer remote-access bleed tool would allow mechanics to bleed the steering rack without removing as much surrounding hardware, saving up to 30 minutes during installation.

Task:

I was responsible for the full design and delivery of the remote-access steering rack bleed tool. This included the CAD, hand calculations, detailed drawings, assembly drawings, supplier sourcing, and manufacturing support. I worked mainly independently, with input from the steering rack designer on the rack requirements and installation process. Since I had limited hydraulics experience at the start of the project, a major part of the task was learning how to design for sealing elements like the spring-ball check valve and O-ring.

Action:

I designed the tool as a quick and easy-to-use remote bleed device with a long extension, sealing O-ring, and integrated spring-ball check valve for steering rack fluid. To size the check valve and sealing features, I performed hand calculations to find the required spring force given a known max pressure, ball size, and valve seat area.

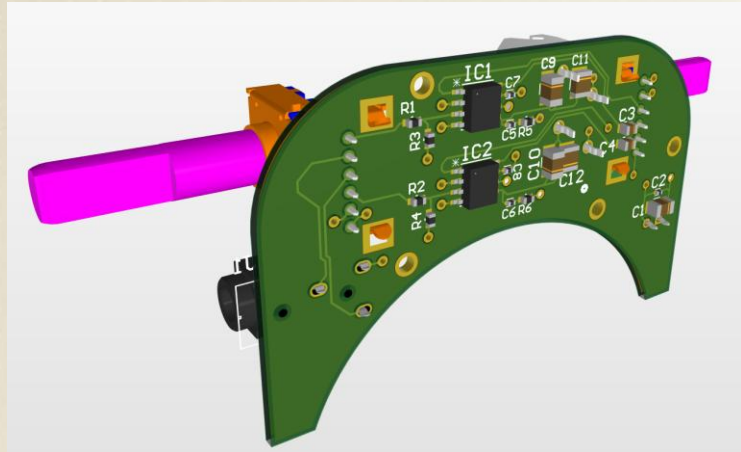
I then developed the CAD model and produced both detailed part drawings and an assembly drawing. I also sourced supplier components and supported manufacturing to make sure the tool could be built and assembled correctly. Throughout the process, I focused on making the tool simple for mechanics to use while still providing the sealing and valve functionality needed for reliable steering rack bleeding with additional handles and flats for a wrench to disassemble the tool during pickup.

Result:

The tool was manufactured and is currently being used. It allows the steering rack to be bled remotely with the nose assembled, reducing assembly time by up to 30 minutes. The project highlighted my ability to take ownership of a practical mechanism-design problem from concept to delivered hardware, quickly learn how to unfamiliar concepts in hydraulic sealing, perform the necessary hand calculations, and deliver a tool that directly improved installation efficiency.

Analog Headphone Personal Project

- Sept. 2024 – Dec. 2024
- Atlanta, GA



Analog Headphone Project (Part 1)

Situation:

I wanted more control over my headphones whenever I was listening to analog audio tracks, especially ones with weak signal outputs and high noise like record players. In addition, I wanted a larger driver with more bass response than current headphones on the market.

Task:

Design a pair of headphones from the ground up to amplify and filter weak and noisy audio signals, which then are played to the user through large drivers.

Action:

I knew that I didn't know how to apply circuits knowledge to the real world at all. So, I began to read up on electrical engineering textbooks, specifically on analog circuits, signal processing, and PCB design. In addition, I extensively consulted my friends in electrical engineering to gain a more intuitive understanding of the process.

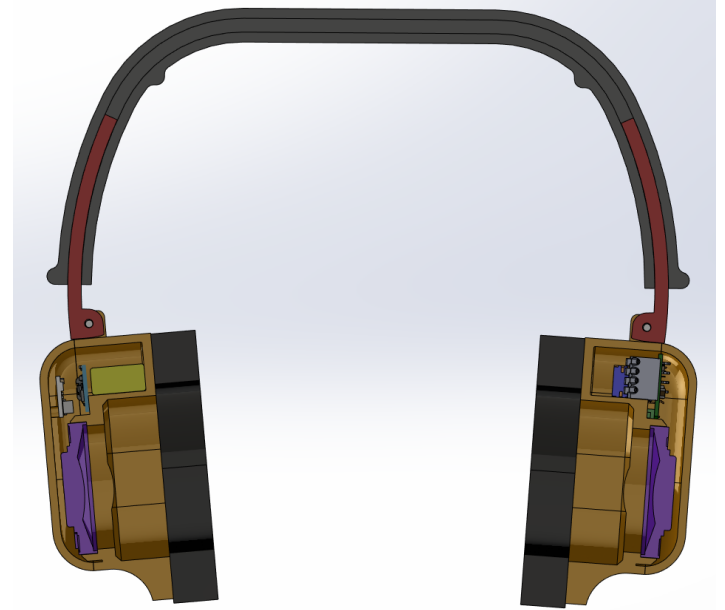
With the knowledge gained, I chose to create a custom amplifying circuit in my headphones. I chose components such as the LM386 op-amp, and designed a schematic and PCB to be the heart of the headphones. In addition, I chose a combination of the 50mm HPD-50N25PR00-32 drivers and an 800 mAh lithium-ion battery, which determined the size of my headphones.

With the electrics done, I focused my attention onto the mechanics of the headphone, which I was much more familiar with. The entire headphone is 3D-printed from PLA and held together with self-tapping M2 and M3 screws to simplify assembly and reduce cost. The earcups have two degrees of freedom, which allow them to extend and rotate. Wire strain relief was accomplished with set screws and a carefully designed headband which prevents the wires from getting caught.

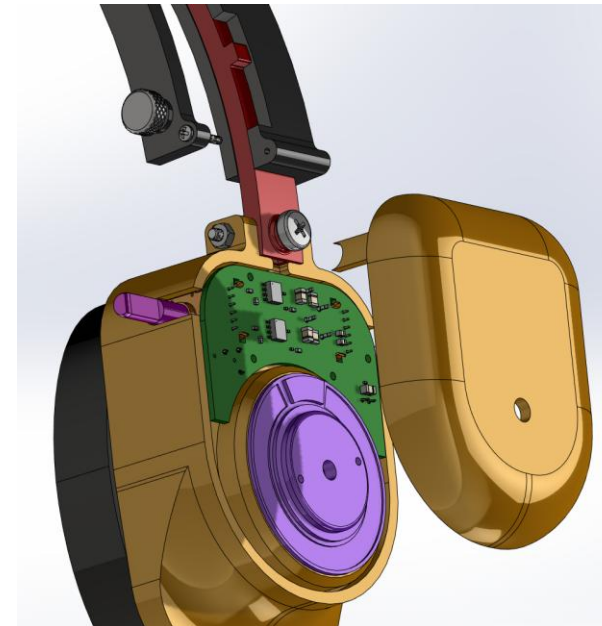
Result:

After many iterations, my circuit successfully played music on a breadboard. The first PCB prototype is currently being manufactured and shipped. As for the headphone mechanics, I am currently in queue for the 3D printers at school, and am waiting on several parts, such as the battery and earpads, to arrive.

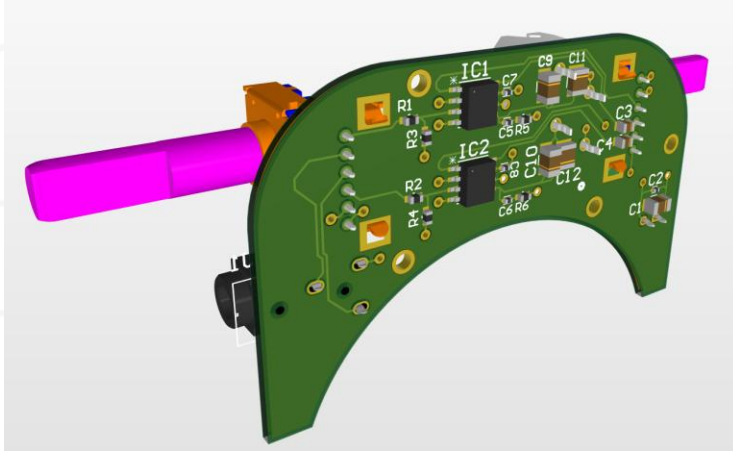
(Pictured Right)
Headphone
Section View



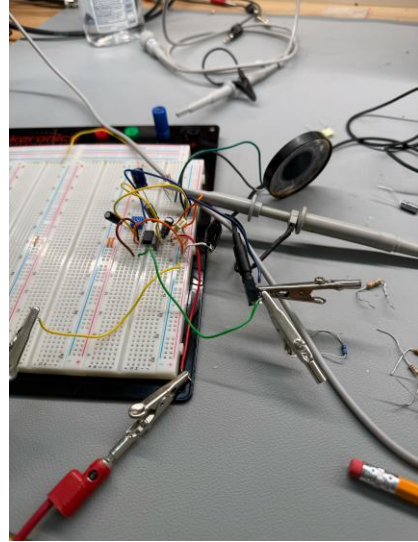
(Pictured Left)
Headphone
Right Earcup
Detailed View



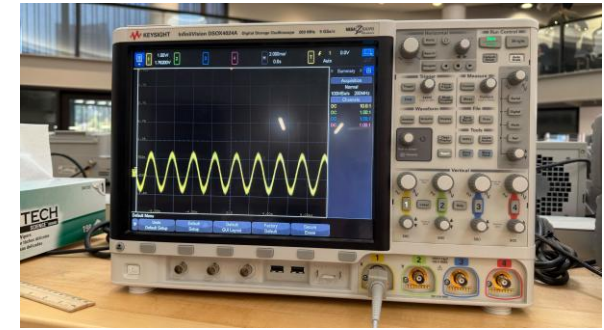
Analog Headphone Project (Part 2)



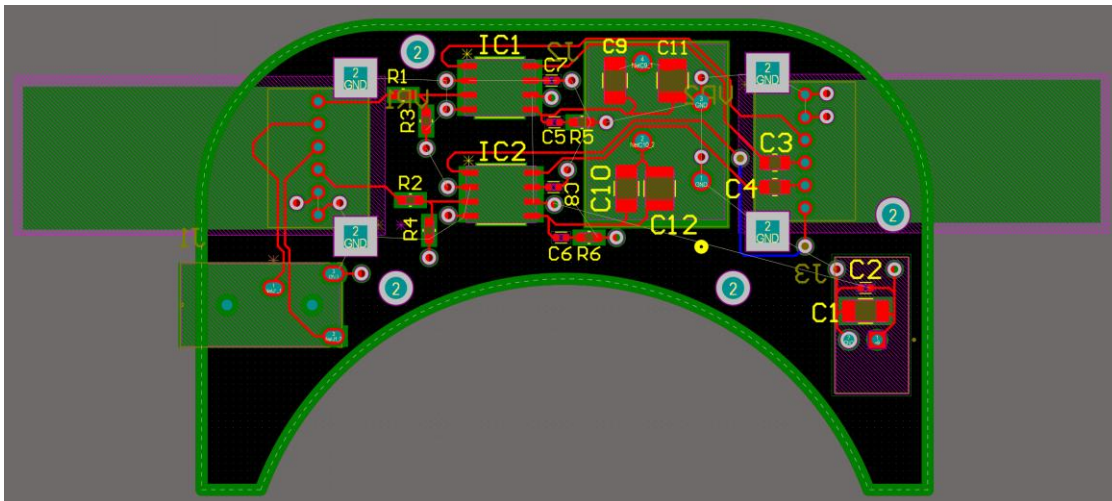
(Pictured Above) PCB Auxiliary View



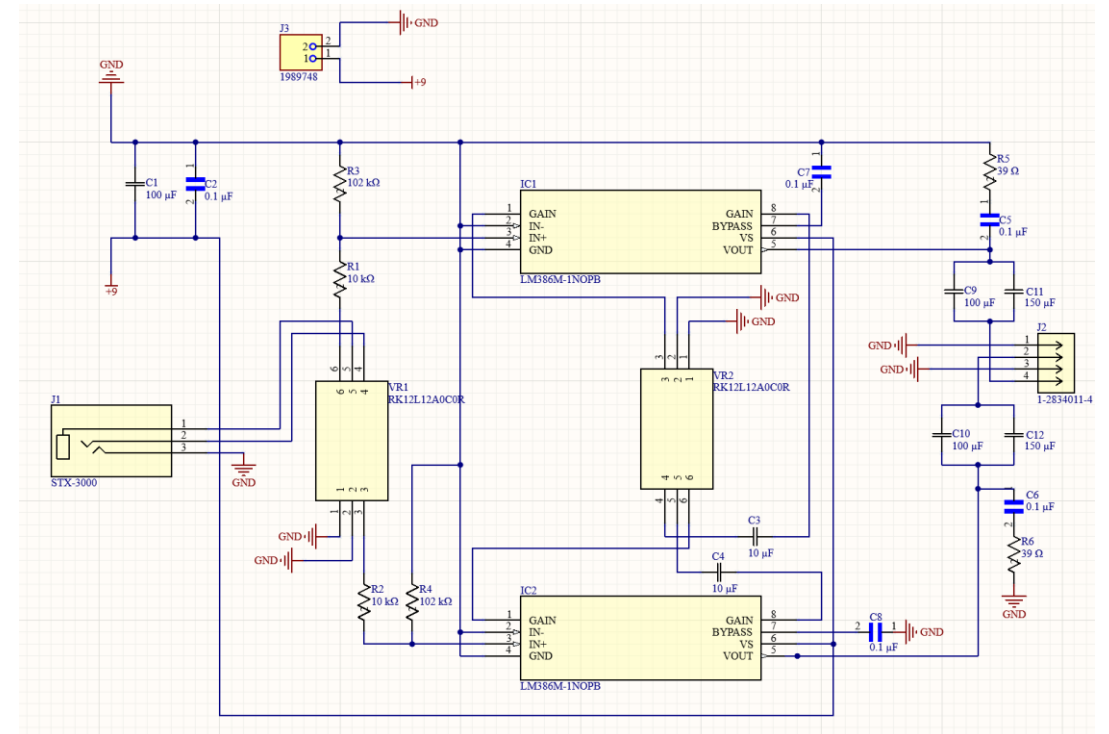
(Pictured Left)
Test Circuit with
Speaker



(Pictured Above) Oscilloscope on output with 20 dB gain



(Pictured Above) PCB Top Layer



(Pictured Above) Circuit Schematic

Tesla Portfolio

- Process & Equipment Engineering Intern
- May 2024 – August 2025
- Reno, NV



Motorized Battery Module Cart Fleet

Situation:

Between two lines of the factory, production associates are moving batteries between machines via forklifts, which increases headcount required for a specific task.

Task:

Create a scalable pushcart solution which can carry up to 1000 kg of batteries safely without exceeding 20 lbs of sustained push force at minimal cost.

Action:

Initially, I investigated re-utilizing older carts for this task with minimal modifications. However, all of the previous carts investigated wouldn't have passed EHS guidelines on push force. Thus, a ground-up design was required. I designed an extruded aluminum frame with a foldable rail and handle system all mounted on 6 caster "see-saw" configuration. The cart balanced on a taller, central caster which significantly reduced initial push force and sustained push force. In addition, existing stocks of materials were used. This design optimized lightness and cost, which is why it lacked more complex features such as a motorized wheel.

However, after testing, the push force failed EHS requirements at 40 lbs of force. I quickly prototyped a motorized version of the cart by negotiating with another team on energy to take on of their older motorized carts. After removing ~400 lbs of excess frame material from the old motorized cart, I adapted it to my new cart's frame, creating a motorized cart with my frame. We achieved EHS approval with this design.

Afterwards I negotiated with the supplier of the old motorized carts to procure custom made ones for our frame, as well as making modifications to the railing system and frame of the new cart.

Result:

The estimated 2-year ROI is \$900,000. Parts for the new motorized cart have been ordered. With a large, rechargeable lithium phosphate battery, it can easily be pushed all day by a single production associate, which will reduce the headcount on over several lines, thus opening new opportunities for increased production elsewhere. In addition, with a yearly maintenance schedule for all of the fasteners, the cart should last more than the desired 2 year lifetime.

Note that the information presented has had all proprietary information removed.
(This includes all photos, videos, and drawings).

Coupler Lubrication Station

Situation:

Currently production associates are manually lubricating battery couplers on a temporary setup (plastic table)

Task:

Create an ergonomic, cheap, and professional coupler lubrication station with enough capacity to supply all battery module lines for 2 years. In addition, the station should have equipment and design for tracking lot numbers and times for lubricated couplers.

Action:

Initially, I designed a low-cost table setup with legs attached directly to the table surface. This setup would have worked in the short-term, with analysis backing up stress results. However, after performing fatigue analysis, it was too flimsy for a 2-year lifespan and was difficult to upgrade in the future. Thus, I redesigned the table using aluminum extrusions and melamine coated MDF. The design features a central printer, HMI, and supply cabinet integrated underneath the main table surface, which is an 2200 x 1100 mm board of melamine coated MDF. There are custom holes drilled for finished coupler buckets to sit flush with the table which improves ergonomics. Finally, the entire table setup is on lockable swivel casters, which makes it versatile in terms of positioning.

Result:

Preventing even one faulty coupler from entering production will pay for the station itself many times over, not to mention the improved rate and storage of lubricated couplers. The significantly improved tracking of couplers improves our capabilities for root cause analysis when diagnosing issues downstream. I've built the station and implemented it into production as of the end of my internship. It is currently in its trial phase of testing with production associates from all of the shifts.

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Coupler Computer Vision Quality Control

Situation:

There is an unacceptable pass rate of mis-assembled battery modules to downstream manufacturing lines and quality, which costs quality teams time and money.

Task:

Train and implement a computer vision model to detect defects at critical junctions of the battery module, ideally with a bias towards false positives.

Action:

I obtained several thousand input images directly from the line, which were already sorted by engineers into passing and failing coupler joints. Luckily, every image was taken in a tightly controlled, high-contrast environment, which simplified training significantly.

These images were then input into a vision model provided by an external supplier, with the default training and learning settings. Given the small amount of training images (constrained due to manually sorting these images), the model was overfit. After several iterations of reducing the learning rate as well as reducing the number of epochs, the model achieved a 95% accuracy with a 5% false positive rate. Any false negative would've been detrimental to quality control, so the model was tuned to have a 0% false negative rate, at the expense of increasing the false positive rate.

Result:

The end result should be a reduction in headcount and improvement in quality, which. This will not only relieve quality teams down the line, but also reduce scrap waste and free up production associates for other lines in need of assistance.

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SpaceX Portfolio

- Starship Mechanisms – Manufacturing Engineering Intern
- Jan. 2024 – May 2024
- Hawthorne, CA



Engineering Master (EM) Process Design

Situation:

Our team just released new TVCA revision. The addition of new features and improvements necessitates a new engineering master (EM) to instruct technicians on how to assemble the TVCA.

Task:

Become the owner of the TVCA manufacturing process by rewriting the current TVCA EM to describe how to assemble the new TVCA, while also improving first-pass yield, cycle time, and clarity. In addition, work with design to adapt to any changing requirements.

Action:

Initially, I was not familiar with the TVCA assembly process and design. So, I immediately set up time to work with technicians to assemble a TVCA unit. Through building an actuator myself, I better understood how the TVCA worked and the difficulties building it. I therefore began rewriting the EM, making sure to include input from stakeholders like the technicians and design engineer on how processes should be designed to reduce incidents of damage or rework due to poor instructions. In addition, I looked into deleting the potting process for the resolver through collaboration with the designer of the TVCA.

Result:

The process has been introduced to the assembly line with clear markings, steps, and pictures describing exactly how to assemble the TVCA unit. I was able to delete a potting step for the resolver, which significantly saved on cycle time. In addition, other improvements like picture documentation replacing CAD views, color coding wires, and detailed measurements has made the assembly process significantly easier, especially for new technicians. This lowers the learning curve for anyone building this unit, which improves first-pass yield.

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Bar-Wound Electric Motor Cutting Process

Situation:

Electric motors from our supplier need post-processing done on some of the bars sticking out. These motors are flight hardware and thus must be processed as quickly as possible while minimizing FOD and ensuring quality.

Task:

Create a repeatable and efficient method to trim motors from supplier to spec for SpaceX welding processes while minimizing FOD and technician time while maximizing remaining material within a short notice.

Action:

I tried using an endmill to trim the stator bars, but this resulted in excess vibration, which led to a poor surface finish and slow process. I designed a 3D-printed jig to hold the bars in place, which improved the vibration but not to the degree desired. Through root-cause analysis, I realized that the endmill's fast feed rate and large teeth caused excessive force to be transmitted to the copper hairpins. An abrasive disk would have been perfect, being able to spin at higher speeds and featuring very small "teeth". However, abrasives embedded themselves in the copper, resulting in poor weld quality. Thus, the solution was to use a high teeth count slit saw mounted onto a high speed Dremel. With everything tightly mounted onto a jig with a rotary table, the copper hairpins were successfully cut.

Result:

This process cuts stators at the expected cycle time of about 30 minutes per stator without impacting laser-weld quality. In addition, target times for the process introduction was met.

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Thrust Vector Control Actuator (TVCA) Build Stand

Situation:

There was an inefficient process to assemble and torque fasteners on Thrust Vector Control Actuator (TVCA) assembly.

Task:

Design a new tool to safely, reliably, and efficiently align, hold, and react on the TVCA to allow for assembly and tightening while also being thin and foldable to increase space for other steps.

Action:

I first visited the technicians performing the process and helped tighten bolts myself. I noted several observations. The TVCA assembly is a cylinder, which is incredibly heavy (>300 lbs) and rotates to allow for access to bolts. The bolts were mounted on a flange which rotates inside of the TVCA assembly (imagine one cylinder rotating in another cylinder). Since bolts need to be tightened star-pattern with a step-up torque, quick rotation of the assembly is critical. This severely limited options to react out torque (>150 ft*lbs) and moments in three dimensions from the assembly.

I first used hand calculations to determine the validity of several designs I drafted, which led me to a hinged wrench design, which would easily slide over the flange and react out torque. Then, in conjunction with the designer, I created removable inserts for the wrench using 304 steel, which had a lower yield strength than the flange so that it would become a sacrificial bearing surfaces which would eventually yield and be replaced. Then, using GD&T, a tolerance stack-up, and CAD, I designed the stand on an aluminum plate with various mounting holes and expansion holes to precisely locate the actuator.

Result:

The new build stand functioned perfectly and was used to assemble the first two prototypes of the revised actuator. No damage was reported on the actuators and the stand did not suffer any noticeable damage. In addition, the cost of the project was lower than expected through the heavy use of 6061 aluminum and internal machine shops.

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Proof and Swage Tool

Situation:

The current proof and swaging tool interferes with geometry on the new TVCA version.

Task:

Design a new proofing and swaging tool that securely installs parts whilst not damaging any flight hardware.

Action:

I observed technicians using the legacy proofing tool. Then, I redesigned the tool to clear the interfering geometry and hand calculated the pressure exerted on flight hardware. Confirming that the tool would not damage any flight hardware, I drafted a drawing, passed checking, and sent the part to be machined.

Result:

The tool has been successfully used on the new TVCA revision without damaging any flight hardware. This project served as a great introduction to SpaceX processes, services, and regulations.

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Rotor Cart

Situation:

There are not enough magnetically safe carts to transport highly magnetic electric motor rotors between assembly areas.

Task:

Design a new cart with a larger batch size to transport highly magnetic rotors between rooms whilst maintaining employee safety.

Action:

I took measurements of the existing cart and got into contact with a foam supplier for SpaceX. After reading up on guidelines for impacts during transportation, I then created a new design for a cart to carry a larger batch size of rotors, with appropriate historical data to back up the design. The cart also did not require much assembly work, which allows it to be easily scalable.

Result:

The new cart cheaply and effectively carries rotors to be moved around in the assembly line.

Note that the information presented has had all proprietary information removed.
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Boeing Portfolio

- Tools & Equipment Engineering Intern – 777/777X Production Engineering
- Everett, WA
- May 2023 – Aug. 2023



Mop Sill Alignment Jig

Situation:

The original mop sill alignment jig tool (~5 lbs.) was dropped onto the aircraft fuselage during production work causing \$5,000 of damage and 19 rework hours.

Task:

A redesign of tool requested to prevent further incident.

Action:

I gathered information about the assembly process from technicians on the floor and read original tool use instructions and assembly instructions. Then, I designed and 3D-printed prototypes of the tool that were significantly lighter (~0.5 lbs.) with filleted edges and significantly cheaper and faster manufacturing. In addition, design also ensured that printing would preserve the incredibly tight tolerances required for alignment. After that, I gave the prototypes to technicians to test on the aircraft during assembly and recorded the feedback. After several design iterations, quotes from industrial engineering, approval from manufacturing engineering, and approval from the Change Control Board III Meeting (CCB3), I drafted and released a technical drawing; the final tool was 3D-printed and sent to technicians.

Result:

The new 3D-printed mop sill alignment jig is more ergonomic, safer, and cheaper than the original tool. It's benefit-cost ratio is relatively high because of how cheap the new tool can be printed. In addition, it can be made completely in-house with zero assembly required, which further reduces costs. My experience completing this project taught me the tooling process, from ideation and design to testing and approval. In addition, I improved on my people skills through collaboration with various teams such as manufacturing engineering, industrial engineering, and technicians. Finally, I learned how to better incorporate reference geometries in models.

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Standard Laser Reflectors

Situation:

Several quality control issues related to drilling had to be addressed because of significant impact to flow and cost. A laser-aided assembly method was proposed, which required standardized reflectors to help align the laser.

Task:

Design standard laser reflectors which fit onto the aircraft.

Action:

This project was a part of a much larger project under my team lead's responsibility. The laser reflectors fit onto the aircraft with powerful magnets into predetermined locations. However, due to the nature of the alignment system, the lasers must have a tight tolerance in order to ensure quality in the assembly process. Thus, the first thing I did was to attend meetings with the outside supplier for the lasers in order to inquire about tolerances, processes, and environmental conditions necessary for operation. I then conducted research into all of the parts and suppliers which needed to be used in the model. Afterwards, I created a CAD model, which was one of the most complex I had done, involving multiple sub-assemblies within assemblies, variations, and reference geometry. I also determined it was necessary to use Geometrical Dimensioning and Tolerancing (GD&T) to create the technical drawing since there were several tight tolerances and it was determined the part was going to be machined. After reading documentation on GD&T standards, I drafted a technical drawing and released it.

Result:

This project required incredible attention to detail and initiative. Keeping track of the various suppliers, working with incredibly tight tolerances, and learning GD&T required intense research and learning. As a result, my capacity to handle more difficult workload increased.

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Drill Exhaust Adapter

Situation:

Technicians encountered significant foreign object debris while drilling holes.

Task:

Building off of previous parts, design a drill adapter to blow away foreign object debris.

Action:

I created a model of the drill exhaust adapter. The 3D printed fitting was already prototyped by another team and was handed to me. Thus, I assembled the model and then began work on the drawing. This required significant research into the Original Equipment Manufacturers (OEMs) which made various parts like the nozzle in order to create a technical drawing. After creating the drawing, it was released.

Result:

The new tool is now ready to use in production. Some of the important lessons I learned from this experience were how to create and manipulate assemblies in CATIA and how to better chase information on OEMs and parts. For the latter learning experience, contacting various people within and outside of the company on information regarding parts and their origins gave me invaluable exposure to the 'messy' parts of engineering. I learned the best ways to find what I wanted, whether it be online, in documentation, or through people.

Note that the information presented has had all proprietary information removed.
(This includes all photos, videos, and drawings).

Seal Paddle

Situation:

Technicians requested a repurpose of a deburring paddle to instead be used with Velcro to smooth seals in the aircraft, as the seal paddle significantly reduced defects in the seals.

Task:

A redesign of tool was requested to improve production quality and flow.

Action:

I went to the shop floor to talk with the technicians and to see for myself some of the seals that were being made. After taking notes and considerations from the technicians, I then performed in-depth research into documentation on what materials were allowed to touch the aircraft and what suppliers and backup suppliers could provide Velcro. Afterwards, I added a new variation of the original deburring tool and updates to the drawing to introduce the new seal paddle into the drawing. This also had the effect of simplifying several steps in the Boeing tooling process, thereby reducing workload and confusion.

Result:

The new seal paddle significantly improves quality in various aircraft seals. These seals are rigorously checked because of their importance to maintaining aircraft pressurization during the thousands pressure cycles that the aircraft will go through during its lifetime. Because of this, having a seal fail quality control causes significant delay to the aircraft production process, which is then passed onto the customer in the form of delays and discounts on the aircraft, which affects the company's profitability. Thus, having a more reliable method of smoothing seals in production can prevent significant delays and discounts later down the line.

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Head Knocker Device

Situation:

Request by technicians to address possible safety hazard due to a sharp protrusion in a cramped environment.

Task:

Create a safety device to prevent injury due to the aforementioned sharp protrusion.

Action:

I began by investigating the protrusion on the shop floor and noting information such as height from foot-level, the material in the sharp protrusion, and several pictures of the area. Using the information gathered from the shop, I was then able to receive a safety score for the tool to be created. The tool was modeled as a conforming 3D-printed pad made of Thermoplastic Polyurethane (TPU) which prevents head injury because of the sharp protrusion. Reference geometry was used in the model. Prototypes were printed and tested, and funding was approved from a CCB3 meeting.

Result:

Although this project only required Computer-Aided Design (CAD) skills with CATIA, it taught me a valuable lesson in understanding the engineering process. To understand a problem and realize a solution, one must gather all the information available about the problem. In this project, understanding the impact of the device and the context by which one could injure themselves in the area significantly aided design work and approval for funding. Thus, by acquiring objective measurements and evidence, the project was able to move at a significantly faster rate than it would have otherwise.

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Foam Kneeling Pad

Situation:

Request by technicians to address safety hazard in cramped environment within the aircraft..

Task:

Create a safety device to prevent injury due to a relatively tall drop-off in the aircraft.

Action:

I started the project by investigating the cramped section of the aircraft where a safety device was requested. There, I managed to climb down myself and take pictures, notes, and measurements. Then, the foam pad was designed with conforming curves to the aircraft, one-way fitment, and stability in mind. In addition, several similar tools were referenced in other aircraft programs to determine firm, fire-retardant foams which could be used. Finally, I reached out to the shop coordinator to contact the vendor who would manufacture the foam pad, which involved long communications about dimensions and material selections.

Result:

Although this project was much smaller than a fully-fledged tool, it still was a significant learning experience. Working with a vendor and customer proved to be much more different than simply producing tools in house, as communication had to be kept clear and concise in order to prevent confusion. This meant that delays, estimated arrival dates, materials, tolerances, and dimensions had to be communicated to two separate parties. As a result, I learned about how to act as a 'middleman' between the vendor and customer and improved upon my people skills.

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Rudder Rig Drill Jig

Situation:

Technicians encountered quality issues with alignment of holes for the rudder.

Task:

A redesign of an existing two-piece tool was assigned.

Action:

This project was intended to provide a prototype for funding approval. The nature of the drill jig meant that the CAD model was incredibly complex, with complex curvatures conforming to the aircraft and consideration of tight tolerances. In addition, the assembly also used many standard screws and parts. In the end, I created a prototype model with many reference geometries.

Result:

My internship ended before the project could be approved. As a result, it is unknown what the impact is. However, I significantly improved on my CATIA CAD modeling skills. I successfully used large and complex reference geometries with a complex assembly. The project proved a great experience in modeling and design.

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Georgia Tech Experimental Rocketry Portfolio

Structures Engineer – Fin Jig Tooling Focal

Atlanta, GA

Aug. 2023 – Aug. 2024



Hypersonic Antenna Shroud

Situation:

It is difficult to remove the external antenna electronics for servicing from the aerodynamic shroud. A new design is required, but one which minimizes its footprint on the rocket, maximizes volume, and has an insignificant impact to aerodynamics.

Task:

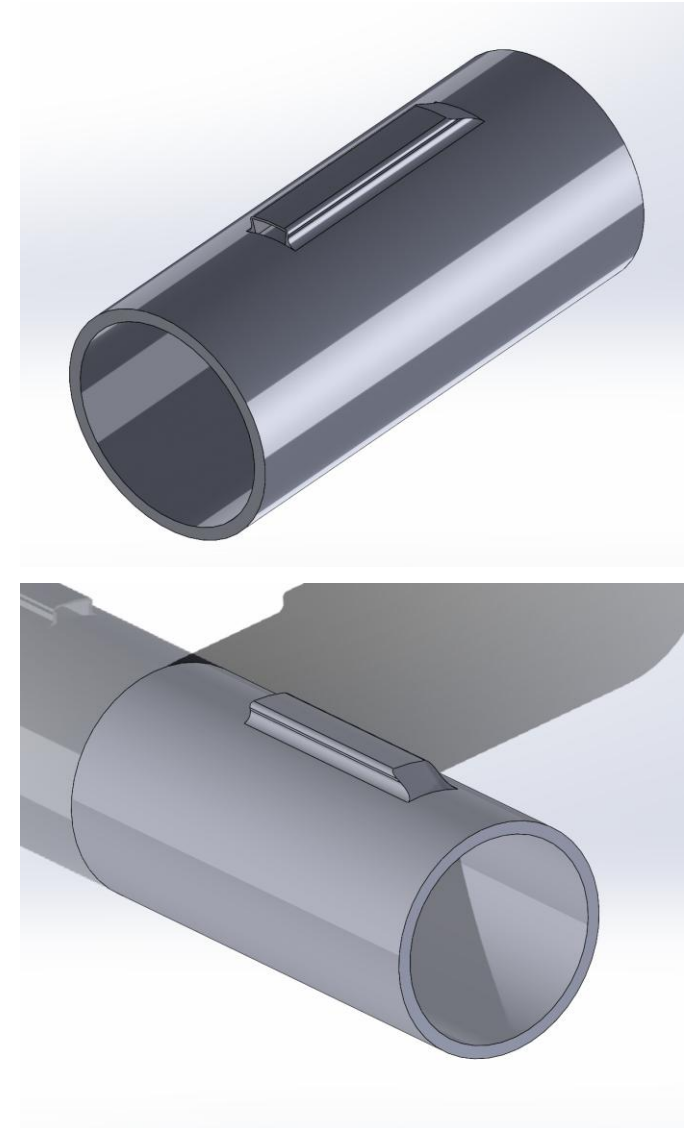
Design a shroud to withstand Mach 4 airflow and temperature stresses while housing all necessary Radio Frequency (RF) antenna electronics. In addition, only RF transparent materials can be used.

Action:

So far, I am currently in the ideation and prototyping phase. The legacy shroud design is a simple conically shaped shroud. This, however, renders a very long section of the shroud's volume unusable. However, using Georgia Tech's paper access, I found a paper discussing the aerodynamic effects of a wedge-shaped shroud on a rocket at hypersonic speeds. Thus, the shroud design I am currently prototyping is based off this research with a specific angle, length, width, and shape to maximize volume while minimizing drag. In addition, the material selection is critical to not only providing RF transparency but also resisting heat.

Result:

Although I do not have a formal education in aerodynamics and thermodynamics just yet, I am excited by the opportunity to learn more about these topics, especially at supersonic and hypersonic flow. Despite only understanding the aerodynamics from a high-level perspective, I believe I can effectively apply a wedge-shaped shroud as a solution to this problem. I am planning on finding someone with expertise in CFD to teach me how to simulate the shroud's aerodynamics to better understand the impact of the wedge shape.



Prototype 1

Fin Alignment Jig

Situation:

There are significant quality control issues with the alignment of rocket fins during the curing process.

Task:

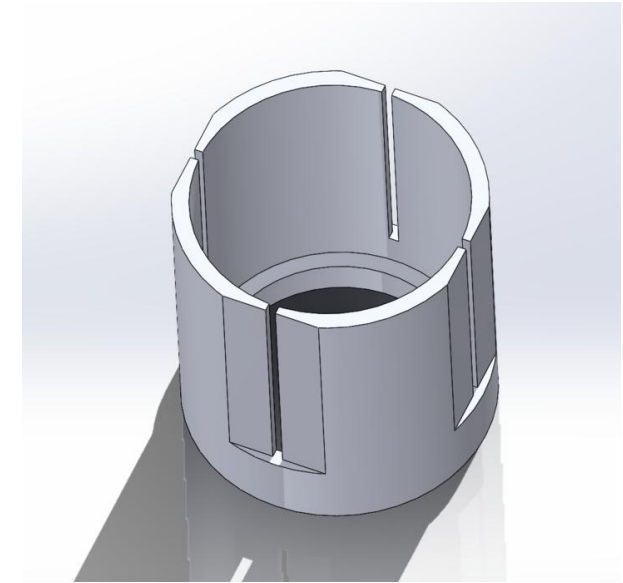
Design an alignment jig to control the fins' spacing, normalcy, and cant while curing at 250 F.

Action:

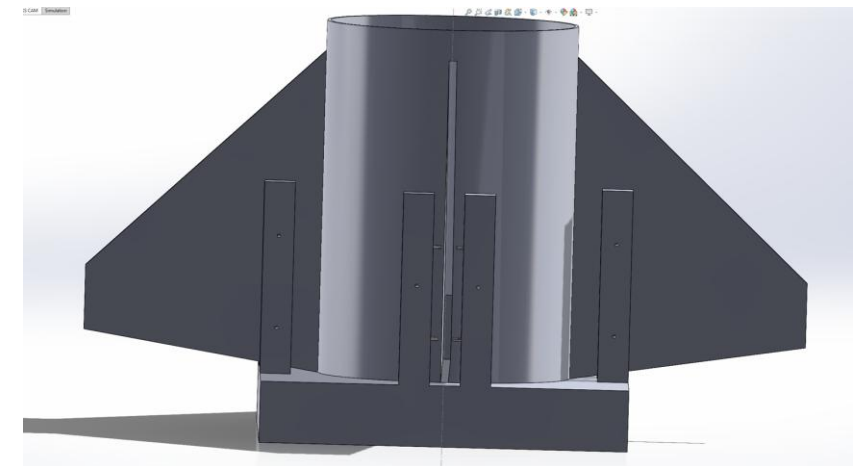
Currently, I am responsible for a team in structures working on designing an alignment jig to hold the fins in place during curing. So far, I have created several prototype jigs and have gathered a significant amount of information on the alignment and curing process. This includes a sample fin, a process document, and a tour of the curing area.

Result:

As I have just begun this role, the only results available are the prototypes which have been designed. However, I have learnt a lot about engineering management, such as providing a direction for the team to follow, dividing up responsibilities, and setting goals.



Prototype 1



Prototype 2

Georgia Tech Motorsports Portfolio

Aerodynamics Engineer

Sep. 2022 – May 2023

Atlanta, GA



Diffuser Layup

Situation:

A new one-piece composite rear diffuser design required intensive manufacturing.

Task:

Cut, assemble and process the mold. Then layup the rear diffuser core and composites.

Action:

I was responsible for assembling the mold pieces to prepare them for machining and the processing the mold afterwards and laying up the composite. The mold was made from Medium Density Fiberboard (MDF) and was machined. During the time-consuming post-processing stage, I used an orbital sander and many grits of sandpaper to achieve a smooth surface on the mold for layups. Afterwards, composite sheets and a Nomex core were laid up onto the mold and sent for curing.

Result:

While this project did not involve much design work, it did provide me with valuable experience in composite manufacturing as well as a rare opportunity to manufacture such a complex part. Attention to detail was also incredibly important during the manufacturing process, which helped improve my engineering skillset.

